



NMAP

GUIDE FOR BEGINNERS

From Zero to Network Scanner

2026 Edition • Written by Rocky

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CHAPTER 01

Introduction to Nmap

■ What is Nmap?

Nmap (Network Mapper) is a free, open-source tool used for network discovery and security auditing. It was designed to rapidly scan large networks, but works fine against single hosts too. Nmap uses raw IP packets in novel ways to determine what hosts are available on the network, what services those hosts are offering, what operating systems they are running, what type of packet filters/firewalls are in use, and dozens of other characteristics.

■ History & Background

Nmap was first published in September 1997 by Gordon Lyon (also known by his pseudonym Fyodor Vaskovich) in a Phrack Magazine article. Since then, it has grown into one of the most widely used network tools in the security community. Today, Nmap is used by system administrators, network engineers, penetration testers, and security researchers worldwide.

■ Why Learn Nmap?

- Identify active hosts on a network quickly
- Discover open ports and running services
- Detect operating systems and software versions
- Audit firewall rules and network security policies
- Perform vulnerability assessments using NSE scripts
- Essential tool for any cybersecurity professional

■ *Note: Always obtain written permission before scanning any network you do not own. Unauthorized scanning may be illegal.*

CHAPTER 02

Installation & Setup

■ Linux Installation

Most Linux distributions include Nmap in their package repositories. Use the appropriate command for your distribution:

```
# Debian/Ubuntu
sudo apt update && sudo apt install nmap

# Red Hat/CentOS/Fedora
sudo dnf install nmap

# Arch Linux
```

```
sudo pacman -S nmap
```

■ Windows Installation

On Windows, download the official installer from the Nmap website (nmap.org). The installer includes Nmap, Zenmap (GUI), Ncat, and Npcap (the packet capture library required on Windows).

```
# After installation, use via Command Prompt or PowerShell:  
nmap --version
```

■ macOS Installation

```
# Using Homebrew (recommended)  
brew install nmap  
  
# Or download the .dmg installer from nmap.org
```

■ Verifying Installation

```
nmap --version  
# Expected output:  
# Nmap version 7.95 ( https://nmap.org )  
# Platform: x86_64-pc-linux-gnu
```

CHAPTER 03

Nmap Basics & Syntax

Basic Command Structure

The general syntax for Nmap is straightforward. Every Nmap command follows this structure:

```
nmap [Scan Type] [Options] {target specification}

# Simple examples:
nmap 192.168.1.1 # Scan a single IP
nmap scanme.nmap.org # Scan a hostname
nmap 192.168.1.0/24 # Scan a subnet
```

Target Specification

Target Type	Example	Description
Single IP	192.168.1.1	Scan one specific host
IP Range	192.168.1.1-254	Scan a range of IPs
CIDR Notation	192.168.1.0/24	Scan entire subnet (256 hosts)
Hostname	example.com	Scan by domain name
From File	-iL targets.txt	Read targets from file
Exclude	--exclude 192.168.1.5	Skip specific hosts

Understanding Output

When Nmap completes a scan, it displays a structured report showing host status, port states, service names, and version information if requested. Here's a sample output breakdown:

```
Nmap scan report for 192.168.1.1
Host is up (0.0023s latency).
PORT STATE SERVICE VERSION
22/tcp open  ssh      OpenSSH 8.9
80/tcp open  http     Apache/2.4
443/tcp open https    Apache/2.4
3306/tcp closed mysql
```

Note: The 'latency' value indicates how long it took to receive a response from the host. Lower values mean a faster/closer host.

CHAPTER 04

Host Discovery

Before scanning ports, Nmap first checks whether the target host is online. This process is called host discovery. By default, Nmap uses multiple techniques to determine if a host is active.

■ Ping Scans

```
# Ping scan only - no port scan
nmap -sn 192.168.1.0/24

# ICMP Echo ping
nmap -PE 192.168.1.1

# TCP SYN ping to port 443
nmap -PS443 192.168.1.1

# UDP ping
nmap -PU53 192.168.1.1
```

■ ARP Scanning

When scanning on a local network (LAN), Nmap automatically uses ARP requests to discover hosts because ARP is more reliable than ICMP on local networks and doesn't rely on firewall rules.

```
# Force ARP scan on local network
nmap -PR 192.168.1.0/24

# ARP scan with verbose output
nmap -PR -v 192.168.1.0/24
```

■ Disabling Ping / Treating All Hosts as Online

```
# Skip host discovery - scan even if host doesn't respond to ping
nmap -Pn 192.168.1.1

# Useful for hosts that block ICMP but have open ports
```

Flag	Discovery Method	Use Case
-sn	Ping Scan (no port scan)	Quick host enumeration
-Pn	No ping (skip discovery)	Firewalled hosts
-PE	ICMP Echo	Standard ping
-PS	TCP SYN ping	When ICMP is blocked
-PA	TCP ACK ping	Alternative to SYN
-PR	ARP ping	Local network scanning

CHAPTER 05

Port Scanning Techniques

Nmap offers a wide variety of scanning techniques, each suited for different scenarios. Choosing the right scan type depends on your goals, network conditions, and permission level.

■ TCP SYN Scan (Stealth Scan)

The default and most popular scan type. It is called a 'stealth' scan because it doesn't complete the TCP handshake, making it less likely to be logged by simple firewalls.

```
# SYN scan (requires root/admin privileges)
sudo nmap -sS 192.168.1.1

# How it works:
# Client sends SYN → Server replies SYN-ACK (open) or RST (closed)
# Client sends RST → Connection never fully established
```

■ TCP Connect Scan

Used when you don't have raw packet privileges. This scan completes the full three-way handshake, making it more detectable but works without elevated privileges.

```
# TCP Connect scan (no root needed)
nmap -sT 192.168.1.1
```

■ UDP Scan

Many important services use UDP (DNS, SNMP, DHCP). UDP scanning is slower than TCP because of the nature of the protocol — there's no handshake to confirm open ports.

```
# UDP scan
sudo nmap -sU 192.168.1.1

# Combine UDP and TCP
sudo nmap -sS -sU 192.168.1.1
```

■ Other Scan Types

Scan Type	Flag	Description
Null Scan	-sN	No flags set; closed ports reply RST
FIN Scan	-sF	Only FIN flag; evades some firewalls
Xmas Scan	-sX	FIN+PSH+URG flags; festive but noisy
ACK Scan	-sA	Maps firewall rules
Window Scan	-sW	Similar to ACK; exploits TCP window

Maimon Scan	-sM	FIN/ACK probe; older technique
SCTP INIT	-sY	For SCTP protocol services

CHAPTER 06

Port Ranges & Port States

■ Specifying Port Ranges

```
# Scan specific port
nmap -p 80 192.168.1.1

# Scan multiple ports
nmap -p 22,80,443,3306 192.168.1.1

# Scan port range
nmap -p 1-1000 192.168.1.1

# Scan all 65535 ports
nmap -p- 192.168.1.1

# Scan top 100 most common ports
nmap --top-ports 100 192.168.1.1

# Scan top 1000 ports (default)
nmap 192.168.1.1
```

■ Port States Explained

State	Meaning
open	A service is actively accepting connections on this port
closed	Port is accessible but no application is listening
filtered	Firewall/filter is blocking Nmap's probes; state unknown
unfiltered	Port is accessible but Nmap can't determine if open/closed
open filtered	Could not determine whether open or filtered (UDP, FIN, etc.)
closed filtered	Cannot determine if closed or filtered

■ *Note: Filtered ports are the most dangerous from a security standpoint — they indicate firewall rules that might be masking active services.*

CHAPTER 07

Service & Version Detection

Knowing that port 80 is open is useful, but knowing it runs Apache 2.4.52 on Ubuntu 22.04 is far more valuable. Nmap's version detection probes open ports and identifies the software listening on them.

Version Scanning

```
# Enable version detection
nmap -sV 192.168.1.1

# Aggressive version detection
nmap -sV --version-intensity 9 192.168.1.1

# Version detection with light probing
nmap -sV --version-intensity 2 192.168.1.1
```

Intensity Levels

Version intensity ranges from 0 (lightest, fastest) to 9 (heaviest, most accurate). The default is 7.

Intensity	Flag	Description
0	--version-intensity 0	Very light, only most likely probes
2	--version-light	Light probes, less network traffic
7	(default)	Balanced accuracy and speed
9	--version-intensity 9	Most aggressive, maximum accuracy
All probes	--version-all	Try every single probe

Note: Higher intensity levels increase accuracy but also increase scan time and network noise. Use higher values only when precision matters.

CHAPTER 08

OS Detection

Nmap can identify the operating system of remote hosts by analyzing subtle differences in TCP/IP stack behavior — a technique known as TCP/IP fingerprinting. Different OSes respond slightly differently to the same network probes.

■ OS Fingerprinting

```
# Enable OS detection
sudo nmap -O 192.168.1.1

# OS detection with version scanning
sudo nmap -O -sV 192.168.1.1

# Limit OS detection attempts
sudo nmap -O --osscan-limit 192.168.1.0/24

# Make OS guesses even if not confident
sudo nmap -O --osscan-guess 192.168.1.1
```

■ Aggressive Detection Mode

The -A flag enables OS detection, version scanning, script scanning, and traceroute all at once. This is the most informative single flag in Nmap.

```
# Aggressive scan (OS + Version + Scripts + Traceroute)
sudo nmap -A 192.168.1.1

# Aggressive scan with verbose output
sudo nmap -A -v 192.168.1.1

# Sample output excerpt:
# OS details: Linux 5.15 (Ubuntu 22.04)
# Network Distance: 1 hop
```

CHAPTER 09

Nmap Scripting Engine (NSE)

The Nmap Scripting Engine (NSE) is one of Nmap's most powerful features. It allows users to write scripts (in Lua) that automate tasks ranging from network discovery to advanced vulnerability detection.

■ What is NSE?

NSE scripts are stored in `/usr/share/nmap/scripts/` (Linux) or the Nmap installation directory (Windows). There are over 600 built-in scripts covering a huge range of tasks. Scripts can interact with target services and return detailed results.

■ Script Categories

Category	Purpose
auth	Test authentication credentials (e.g., default passwords)
broadcast	Discover hosts not in target list via broadcast
brute	Brute-force credential attacks
default	Run with <code>-sC</code> ; safe scripts providing useful info
discovery	Query public registries, SNMP, DNS, etc.
dos	Denial-of-service tests (use with caution!)
exploit	Exploit known vulnerabilities
external	Interact with external resources (Whois, DNS)
fuzzer	Send fuzzing payloads to detect bugs
intrusive	Risky scripts likely to cause issues
malware	Test for backdoors and malware infections
safe	Low-risk scripts, unlikely to crash services
version	Help identify service versions
vuln	Check for known CVE vulnerabilities

■ Running Scripts

```
# Run default scripts
nmap -sC 192.168.1.1

# Run specific script
nmap --script=http-title 192.168.1.1

# Run multiple scripts
nmap --script=http-title,http-headers 192.168.1.1
```

```
# Run all scripts in a category
nmap --script=vuln 192.168.1.1

# Run script with arguments
nmap --script=http-brute --script-args userdb=users.txt 192.168.1.1

# Update script database
sudo nmap --script-updatedb
```

■ Popular Scripts

Script	Description
http-title	Fetches the title of web pages
ssh-brute	Brute-force SSH logins
smb-vuln-ms17-010	Detect EternalBlue (WannaCry) vulnerability
ssl-heartbleed	Test for Heartbleed SSL vulnerability
ftp-anon	Check if FTP allows anonymous login
dns-brute	Brute-force DNS subdomains
mysql-empty-password	Check MySQL for empty root password
http-robots.txt	Fetch and display robots.txt file
banner	Simple service banner grabbing
whois-domain	Perform WHOIS lookup on target domain

CHAPTER 10

Output & Reporting

Nmap supports several output formats, making it easy to save scan results for later analysis, reporting, or processing by other tools.

■ Output Formats

Format	Flag	Description
Normal	-oN scan.txt	Human-readable text output
XML	-oX scan.xml	Machine-parseable XML; importable by tools
Grepable	-oG scan.gnmap	Easily parsed with grep/awk
Script Kiddie	-oS scan.txt	l33t-sp34k format (fun, not useful!)
All formats	-oA scan	Save in all formats simultaneously

■ Saving Results

```
# Save as normal text
nmap -sV -oN results.txt 192.168.1.0/24

# Save as XML for Metasploit import
nmap -sV -oX results.xml 192.168.1.0/24

# Save all formats at once
nmap -sV -oA full_scan 192.168.1.0/24
# Creates: full_scan.nmap, full_scan.xml, full_scan.gnmap

# Append to existing file
nmap -sV --append-output -oN results.txt 192.168.1.5

# View XML results with browser or parse with Python/Ruby
```

CHAPTER 11

Timing & Performance

Nmap's timing options control how fast or slow scans run. Faster scans are louder on the network (easier to detect), while slower scans are stealthier but take longer.

■ Timing Templates

Template	Flag	Name	Use Case
T0	-T0	Paranoid	Extremely slow; IDS evasion
T1	-T1	Sneaky	Slow; avoids most IDS
T2	-T2	Polite	Slower; reduces bandwidth usage
T3	-T3	Normal	Default speed
T4	-T4	Aggressive	Fast; assumes reliable network
T5	-T5	Insane	Very fast; may miss results

■ Fine-Tuning Performance

```
# Set maximum parallel port probes
nmap --min-parallelism 10 --max-parallelism 100 192.168.1.0/24

# Set host timeout (abandon slow hosts)
nmap --host-timeout 30s 192.168.1.0/24

# Set probe timeout
nmap --min-rtt-timeout 100ms --max-rtt-timeout 1000ms 192.168.1.1

# Limit scan rate (packets per second)
nmap --min-rate 100 --max-rate 500 192.168.1.0/24
```

■ *Note: For production network scanning, T3 (default) or T2 (polite) are safest. Use T4 only on isolated test networks.*

CHAPTER 12

Firewall & IDS Evasion

When conducting authorized penetration tests, you may need to evade firewalls or intrusion detection systems. Nmap provides several techniques for this purpose.

■ Packet Fragmentation

Splitting packets into tiny fragments can confuse some simple firewalls and packet filters.

```
# Fragment packets into 8-byte chunks
sudo nmap -f 192.168.1.1

# Fragment into 16-byte chunks
sudo nmap -ff 192.168.1.1

# Specify fragment size manually
sudo nmap --mtu 16 192.168.1.1
```

■ Decoy Scanning

Make your scan appear to come from multiple IP addresses simultaneously, hiding your real source in the noise.

```
# Use decoy IPs (ME = your real IP position)
sudo nmap -D 10.0.0.1,10.0.0.2,ME 192.168.1.1

# Generate random decoys
sudo nmap -D RND:10 192.168.1.1
```

■ Source IP Spoofing & Port

```
# Spoof source IP (you won't receive replies)
sudo nmap -S 1.2.3.4 192.168.1.1

# Use specific source port (e.g., 53 = DNS, often whitelisted)
sudo nmap --source-port 53 192.168.1.1
sudo nmap -g 80 192.168.1.1

# Spoof MAC address
sudo nmap --spoof-mac 0 192.168.1.1 # Random MAC
sudo nmap --spoof-mac Cisco 192.168.1.1 # Vendor MAC
```

■ *Note: All evasion techniques should only be used during authorized penetration testing or on networks you own.*

CHAPTER 13

Zenmap – The Nmap GUI

Zenmap is the official graphical user interface for Nmap. It is cross-platform, free, and is bundled with the Windows and macOS Nmap installers. Zenmap makes Nmap accessible to beginners while still providing full access to advanced features.

■ Key Features of Zenmap

- Profile-based scanning with pre-configured scan types
- Interactive topology map showing network relationships
- Side-by-side scan comparison for detecting changes
- Searchable results with powerful filtering
- Save and load scan results for later analysis
- Host list with sortable columns and detailed info panels

■ Predefined Scan Profiles

Profile Name	Command Used	Purpose
Intense scan	<code>nmap -T4 -A -v</code>	Most info, aggressive
Intense scan + UDP	<code>nmap -sS -sU -T4 -A -v</code>	TCP + UDP scan
Ping scan	<code>nmap -sn</code>	Discover live hosts only
Quick scan	<code>nmap -T4 -F</code>	Fast top-100 port scan
Quick traceroute	<code>nmap -sn --traceroute</code>	Map network path
Regular scan	<code>nmap</code>	Default scan
Slow comprehensive	<code>nmap -sS -sU -T4 -A -v -PE --script discovery,safer</code>	Safe audit

CHAPTER 14

Real-World Use Cases

■ Network Inventory & Auditing

```
# Discover all live hosts on the network
nmap -sn 192.168.1.0/24 -oN live_hosts.txt

# Full inventory scan
sudo nmap -sS -sV -O -T4 192.168.1.0/24 -oA network_audit

# Find all web servers
nmap -p 80,443,8080,8443 --open 192.168.1.0/24

# Find all SSH servers
nmap -p 22 --open 192.168.1.0/24
```

■ Vulnerability Assessment

```
# Check for common vulnerabilities
sudo nmap --script=vuln 192.168.1.1

# Check for EternalBlue (MS17-010)
nmap --script smb-vuln-ms17-010 -p445 192.168.1.1

# Check SSL/TLS issues
nmap --script ssl-enum-ciphers -p 443 192.168.1.1

# HTTP security headers check
nmap --script http-security-headers -p 80,443 192.168.1.1

# Check for anonymous FTP
nmap --script ftp-anon -p 21 192.168.1.0/24
```

■ Service Enumeration

```
# Database servers
nmap -p 1433,3306,5432,27017 --open 192.168.1.0/24

# SMB / Windows shares
nmap -p 445 --script smb-enum-shares 192.168.1.1

# SNMP enumeration
sudo nmap -sU -p 161 --script snmp-info 192.168.1.0/24

# DNS zone transfer attempt
nmap --script dns-zone-transfer -p 53 192.168.1.1
```

CHAPTER 15

Nmap Cheat Sheet

A quick-reference summary of the most commonly used Nmap commands and flags for everyday use.

Scan Types

Command	Description
nmap -sS <target>	TCP SYN Scan (Stealth)
nmap -sT <target>	TCP Connect Scan
nmap -sU <target>	UDP Scan
nmap -sN <target>	Null Scan
nmap -sF <target>	FIN Scan
nmap -sX <target>	Xmas Scan
nmap -sA <target>	ACK Scan (firewall mapping)
nmap -sn <target>	Ping Scan (no ports)

Discovery & Ports

Command	Description
nmap -Pn <target>	Skip ping, treat as online
nmap -p 80 <target>	Scan port 80
nmap -p 1-1000 <target>	Scan port range
nmap -p- <target>	Scan all 65535 ports
nmap --top-ports 100 <target>	Scan top 100 ports
nmap --open <target>	Show only open ports

Detection & Info

Command	Description
nmap -sV <target>	Version detection
nmap -O <target>	OS detection
nmap -A <target>	Aggressive (OS+Ver+Script+Trace)
nmap -sC <target>	Default script scan
nmap --script=vuln <target>	Vulnerability scripts
nmap --script=<name> <target>	Run specific NSE script

Output & Timing

Command	Description
<code>nmap -oN file.txt <target></code>	Save normal output
<code>nmap -oX file.xml <target></code>	Save XML output
<code>nmap -oA scan <target></code>	Save all formats
<code>nmap -v <target></code>	Verbose output
<code>nmap -vv <target></code>	Very verbose
<code>nmap -T4 <target></code>	Aggressive timing
<code>nmap -T2 <target></code>	Polite timing

Evasion & Misc

Command	Description
<code>nmap -f <target></code>	Fragment packets
<code>nmap -D RND:10 <target></code>	Random decoys
<code>nmap -g 53 <target></code>	Use port 53 as source
<code>nmap --spoof-mac 0 <target></code>	Random MAC address
<code>nmap -iL hosts.txt</code>	Input from file
<code>nmap --exclude 192.168.1.5 <target></code>	Exclude a host
<code>nmap --reason <target></code>	Show reason for port state
<code>nmap --traceroute <target></code>	Include traceroute

Legal Notice & Conclusion

Nmap is an incredibly powerful tool that can be used for both offensive and defensive purposes. As with any security tool, ethical and legal use is absolutely essential.

■ Legal & Ethical Reminder

■ IMPORTANT LEGAL NOTICE

Scanning networks, systems, or devices without explicit written authorization from the owner is illegal in most countries and may result in criminal prosecution. Always obtain written permission before scanning any network. Nmap was designed for authorized network administration and security auditing. Use it responsibly and ethically at all times.

■ Conclusion

You have now covered the essential foundations of Nmap — from basic host discovery and port scanning to advanced NSE scripting and evasion techniques. Nmap remains one of the most versatile and widely-used tools in any security professional's toolkit.

Continue your learning journey by:

- Practicing on intentionally vulnerable systems like Metasploitable, HackTheBox, or TryHackMe
- Exploring the full Nmap documentation at nmap.org/book
- Writing custom NSE scripts in Lua
- Combining Nmap output with tools like Metasploit, OpenVAS, and Nessus
- Setting up a home lab for ethical hacking practice

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